MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING TYPE 2N560

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a NPN, silicon, low-power, switching transistor.

* 1.2 Physical dimensions. See figure 1 (TO-5).

* 1.3 Maximum ratings.

P_T 1/ T_A = 25°C	V_{CBO}	V_{CES}	V_{EBO}	I_C	T_J	T_{stg}
mW	Vdc	Vdc	Vdc	mAdc	°C	°C
500	60	60	8	100	+200	-65 to +200

1/ Derate linearly 2.85 mW/°C for $T_A > 25°C$.

* 1.4 Primary electrical characteristics.

Limits	h_{FE}	C_{obo}	V_{CE}(sat)	V_{BE}(sat)	Switching	
	V_{CE} = 5 Vdc I_C = 100 mAdc	V_{CB} = 5 Vdc I_E = 0 100 kHz ≤ f ≤ 1 MHz	I_C = 10 mAdc I_B = 0.4 mAdc	IC = 10 mAdc I_B = 0.4 mAdc	t_{on}	t_{off}
		pf	Vdc	Vdc	nsec	nsec
Min	20	---	---	0.7	---	---
Max	200	8	0.55	0.9	60	250

FSC 5961

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2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

* 3.3.1 Terminal-lead length. Terminal-lead length(s) other than that specified in figure 1 may be furnished when so stipulated under contract or order (see 6.2) where the device covered herein is required directly for particular equipment-circuit installation or for automatic-assembly-technique programs. Where other lead lengths are required and provided, it shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking.

* 3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

* 3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

* 4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

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* 4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A and B inspections. When specified in the contract or order, one copy of the quality conformance inspection data, pertinent to the device inspection lot, shall be supplied with each shipment by the device manufacturer (see 6.2).

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.4 Methods of examination and test. *Methods of examination and test shall be as specified in* tables I and II, and as follows:

* 4.4.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
* Subgroup 2			5				
Breakdown voltage, collector to emitter	3011	Bias cond. C; I_C = 10 μAdc		BV_{CES}	60	---	Vdc
Breakdown voltage, collector to base	3001	Bias cond. D; I_C = 10 μAdc		BV_{CBO}	60	---	Vdc
Breakdown voltage, emitter to base	3026	Bias cond. D; I_E = 10 μAdc		BV_{EBO}	8	---	Vdc
Collector to base cutoff current	3041	Bias cond. D; V_{CB} = 20 Vdc		I_{CBO}	---	50	nAdc
* Subgroup 3			5				
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 100 mAdc; pulsed (see 4.4.1)		h_{FE}	20	200	---
Base emitter voltage (saturated)	3066	Test cond. A; I_C = 10 mAdc; I_B = 0.4 mAdc		$V_{BE}(sat)$	0.7	0.9	Vdc
Collector to emitter voltage (saturated)	3071	I_C = 10 mAdc; I_B = 0.4 mAdc		$V_{CE}(sat)$	---	0.55	Vdc

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TABLE I. Group A inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 4			5				
Open circuit output capacitance	3236	V_{CB} = 5 Vdc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz		C_{obo}	---	8	pf
Pulse response	3251	Test cond. A					
Turn-on time	---	R_C = 500 ohms; V_{CC} = 5 Vdc; $I_{B(0)}$ = 0; $I_{B(1)}$ = 1 mAdc		t_{on}	---	60	nsec
Turn-off time	---	R_C = 500 ohms; V_{CC} = 5 Vdc; $I_{B(1)}$ = 0.5 mAdc; $I_{B(2)}$ = -0.5 mAdc		t_{off}	---	250	nsec

TABLE II. Group B inspection.

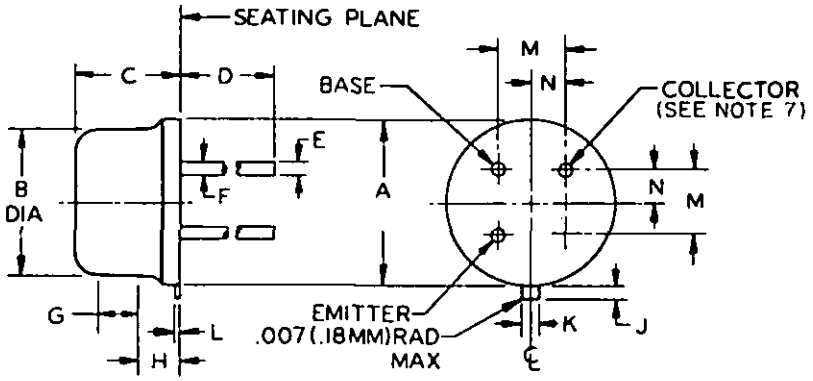
Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
* Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A		---	---	---	---
Seal (leak-rate)	---	MIL-STD-202, method 112, test cond. C, procedure III; test cond. A for gross leaks		---	---	$1x10^{-7}$	atm cc/sec
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 20 Vdc		I_{CBO}	---	50	nAdc
Collector to emitter voltage (saturated)	3071	I_C = 10 mAdc; I_B = 0.4 mAdc		$V_{CE(sat)}$	---	0.55	Vdc
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 100 mAdc; pulsed (see 4.4.1)		h_{FE}	20	200	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 3			15				
Shock	2016	Nonoperating; 1500 G; 0.5 msec; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration fatigue	2046	Nonoperating		---	---	---	---
Vibration, variable frequency	2056	Nonoperating		---	---	---	---
Constant acceleration	2006	20,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
* Subgroup 5			15				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points: (Same as subgroup 2)							
* Subgroup 6			$\lambda = 10$				
High-temperature life (nonoperating)	1031	T_{stg} = +200°C		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = 20 Vdc		I_{CBO}	---	100	nAdc
Forward-current transfer ratio	3076	V_{CE} = 5 Vdc; I_C = 100 mAdc; pulsed (see 4.4.1)		Δh_{FE}	---	±25	% change from initial Group A reading
* Subgroup 7			$\lambda = 10$				
Steady-state operation life	1026	T_A = 25°C; V_{CB} = 10 Vdc; P_T = 500 mW		---	---	---	---
End points: (Same as subgroup 6)							

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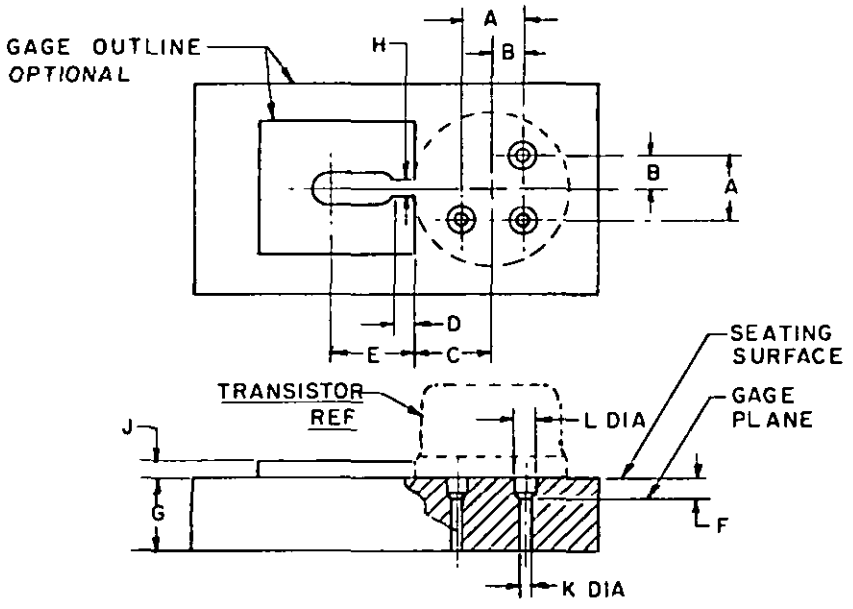


DIMENSIONS					NOTES
LTR	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.335	.370	8.51	9.40	
B	.305	.335	7.75	8.51	
C	.240	.260	6.10	6.60	
D	1.500	1.750	38.10	44.45	9
E	.016	.021	.41	.53	2,9
F	.016	.019	.41	.48	3,9
G	.100	---	2.54	---	4
H	---	---	---	---	5
J	.029	.045	.74	1.14	8
K	.028	.034	.71	.86	
L	.009	.125	.23	3.18	
M	.1414 Nom		3.59 Nom		6
N	.0707 Nom		1.80 Nom		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. Variations on dimension B in this zone shall not exceed .010 (.25 mm).
5. Outline in this zone is not controlled.
6. When measured in a gaging plane .054+.001, −.000 (1.37+.03, −.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 2 shows the preferred measured method.
7. The collector shall be internally connected to the case.
8. Measured from the maximum diameter of the actual device.
9. All 3 leads. (see 3.3.1).

* FIGURE 1. Physical dimensions of transistor type 2N560 (TO-5).

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NOTES:
1. The following gaging procedure shall be used: The use of a pin straightener prior to insertion in the gage is permissible. The device being measured shall be inserted until its seating plane is .125±.010 (3.18±.25 mm) from the seating surface of the gage. A spacer may be used to obtain the .125 (3.18 mm) distance from the gage seat prior to force application. A force of 8 oz± .5 oz shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.
2. The location of the tab locator, within the limits of *dimension C, will be determined* by the tab and flange dimension of the device being checked.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.1409	.1419	3.58	3.60
B	.0702	.0712	1.78	1.81
C	.182	.199	4.62	5.05
D	.009	.011	.23	.28
E	.125 Nom		3.18 Nom	
F	.054	.055	1.37	1.40
G	.372	.378	9.45	9.60
H	.0350	.0355	.89	.90
J	.150 Nom		3.81 Nom	
K	.0325	.0335	.83	.85
L	.0595	.0605	1.51	1.54

* FIGURE 2. Gage for lead and tab location for transistor type 2N560.

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5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Procurement document should specify:

(a) Terminal-lead length (see 3.3.1).
(b) Inspection data (see 4.3).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - EL
Navy - EC
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Review activities:
Army - EL, MU, MI
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army- EL, SM
Navy - CG, MC, AS, OS
Air Force - 19

Preparing activity:
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